NCA Exercises

Ex. 1

Larken Ltd purchased factory equipment with an invoice price of $50,000. Other costs incurred were freight costs, $1,300; installation wiring and foundation, $2,200; material and labor costs in testing equipment, $700; oil lubricants and supplies to be used during the year with equipment, $500; annual fire insurance policy covering equipment, $1,400. The equipment is estimated to have a $5,000 salvage value at the end of its 8-year useful service life.

Instructions

(a) Calculate the acquisition cost of the equipment. Clearly identify each element of cost.

(b) If the straight-line method of depreciation was used, the annual rate applied to the depreciable cost would be __________.

Solution 1 *(10 min.)*

(a)

Invoice cost	$50,000
Freight costs	1,300
Installation wiring and foundation	2,200
Material and labor costs in testing	700
Acquisition cost	$54,200

(b) If the straight-line method of depreciation was used, the annual rate applied to the depreciable cost would be 12.5% (100% ÷ 8 years).

Q) Chip Manufacturing purchases property that includes land, buildings and equipment for $2.1 million. The company also pays $19,000 in legal fees, $21,000 in commissions, and $10,000 in appraisal fees. The fair value of the property is allocated as follows land is estimated at 50%, the buildings are at 30%, and the equipment at 20% of the property value.
Required:

a. Determine the total acquisition cost of this "basket purchase". **(Enter your answer in whole dollars and not in millions.)**

b. Allocate the total acquisition cost to the individual assets acquired. **(Enter your answers in whole dollars and not in millions.)**

Q)
At the beginning of 2021, your company buys a $46,400 piece of equipment that it expects to use for 4 years. The equipment has an estimated residual value of 6,400. The company expects to produce a total of 500,000 units. Actual production is as follows: 110,000 units in 2021, 115,000 units in 2022, 127,000 units in 2023, and 18,000 units in 2024.

Required:

a. Determine the depreciable cost.

b. Calculate the depreciation expense per year under the straight-line method.

c. Use the straight-line method to prepare a depreciation schedule.

d. Calculate the depreciation rate per unit under the units-of-production method. **(Round your answer to 2 decimal places**.)

e. Use the units-of-production method to prepare a depreciation schedule. **(Do not round your Depreciation rate per unit.)**

Ex 26

Pine Ltd purchased a truck for $40,000. The company expected the truck to last four years or 100,000 miles, with an estimated residual value of $4,000 at the end of that time. During the second year the truck was driven 27,500 miles. Calculate the depreciation for the second year under each of the methods below and place your answers in the blanks provided.

Straight-line	$__________________
Units-of-activity	$__________________
Diminishing balance at 25% p.a.	$__________________

Solution 26 (10 min.)

Straight-line $ 9,000
(40,000 - 4,000/4 = 9,000)

Units-of-activity $ 9,900
(40,000 - 4,000/100,000*27,500 = 9,900)

Diminishing balance at 25% p.a. $ 7 500
(year 1 — 40,000 x .25 = 10,000)
(year 2 — (40,000 - 10,000) x .25 = 10,000)

Ex. 27

The Tolbert Corporation purchased equipment on 1 January, 2018, for $60,000. It is estimated that the equipment will have a $5,000 salvage value at the end of its 5-year useful life. It is also estimated that the equipment will produce 100,000 units over its 5-year life.

Instructions
Answer the following independent questions.

1. Calculate the amount of depreciation expense for the year ended 31 December, 2018, using the straight-line method of depreciation.
2. If 16,000 units of product are produced in 2018 and 24,000 units are produced in 2019, what is the book value of the equipment at 31 December, 2019? The company uses the units-of-activity depreciation method.

Solution 27 *(15 min.)*

$$\text{1. Straight-line method: } \frac{\text{C - S}}{\text{Years}} = \frac{\$60{,}000 - \$5{,}000}{5} = \$11{,}000 \text{ per year.}$$

$$\text{2. Units-of-activity method: } \frac{\text{C - S}}{\text{Units}} = \frac{\$60{,}000 - \$5{,}000}{100{,}000 \text{ units}} = \$.55 \text{ per unit}$$

2008	16,000 units x $.55	=	$ 8,800
2009	24,000 units x $.55	=	13,200
Accumulated Depreciation		=	$22,000

Cost of asset	$60,000
Less: Accumulated Depreciation	22,000
Book value at 31 December, 2009	$38,000

Ex. 28

A plant asset acquired on October 1, 2019, at a cost of $400,000 has an estimated useful life of 10 years. The salvage value is estimated to be $30,000 at the end of the asset's useful life.

Instructions
Determine the depreciation expense for the first two years using the:
(a) straight-line method.
(b) double-declining-balance method.

Solution 28 *(5 min.)*

(a) Straight-line method

Year 1 $\frac{\$400{,}000 - \$30{,}000}{10 \text{ years}} = \$37{,}000 \text{ x } 3 \div 12 = \underline{\$9.250}$

Year 2 $37,000

(b) Double-declining-balance method

Constant rate — 2 / 10 = 20%

Year 1 $400,000 x 20% x 3 / 12 = $20,000

Year 2 $380,000 x 20% = $76,000

Ex. 30

The Donovan Medical Centre purchased a new surgical laser for \$64,000. The estimated salvage value is \$4,000. The laser has a useful life of six years and the clinic expects to use it 10,000 hours. It was used 1,600 hours in year 1; 2,100 hours in year 2; 2,400 hours in year 3; 1,400 hours in year 4; 2,000 hours in year 5; 500 hours in year 6.

Instructions

(a) Calculate the annual depreciation for each of the six years under each of the following methods:
   (1) straight-line.
   (2) units-of-activity.

(b) If you were the administrator of the clinic, which method would you deem as most appropriate? Justify your answer.

(c) Which method would result in the lowest reported income in the first year? Which method would result in the lowest total reported income over the six-year period?

Solution 30 *(15 min.)*

(a) (1) Straight-line method: $\frac{\$64{,}000 - \$4{,}000}{6 \text{ years}} = \$10{,}000 \text{ per year}$

(2) Units-of-activity method: $\frac{\$64{,}000 - \$4{,}000}{10{,}000 \text{ hours}} = \$6.00 / \text{hour}$

Year	1	1,600	x	\$6.00	=	\$ 9,600
	2	2,100	x	6.00	=	12,600
	3	2,400	x	6.00	=	14,400
	4	1,400	x	6.00	=	8,400
	5	2,000	x	6.00	=	12,000
	6	500	x	6.00	=	3,000

	Straight-line	Units of Activity
Year 1	\$10,000	\$ 9,600
Year 2	10,000	12,600
Year 3	10,000	14,400
Year 4	10,000	8,400
Year 5	10,000	12,000
Year 6	10,000	3,000
Total	\$60,000	\$60,000

(b) The units-of-activity method can be justified based on the variable usage the laser will receive during its useful life.

(c) The straight-line method provides the highest depreciation expense for the first year, and therefore the lowest first year income. Over the six-year period, both methods result in the same total depreciation expense (\$60,000) and, therefore, the same total income.

Ex. 6

Radiata Ltd purchased equipment on 1 January, 2017, at a cost of $60,000. The equipment was originally estimated to have a salvage value of $5,000 and an estimated life of 10 years. Depreciation has been recorded through 31 December, 2020, using the straight-line method. On 1 January, 2020, the estimated salvage value was revised to $6,000 and the useful life was revised to a total of 8 years.

Instructions

Determine Radiata Ltd's Depreciation Expense for 2020.

Solution 6 *(5 min.)*

Calculate the Book Value at the time of the revision:

$$\frac{\$60{,}000 - \$5{,}000}{10 \text{ years}} = \$5{,}500 \text{ annual depreciation expense}$$

4 years have been depreciated: $5,500 x 4 = $22,000

Book Value at the time of the revision: $60,000 - $22,000 = $38,000

Calculate the revised annual depreciation:

$$\frac{\$38{,}000 - \$6{,}000}{4 \text{ years remaining}} = \$8{,}000 \text{ revised annual depreciation}$$

The Depreciation Expense for 2010 is $8,000.

Ex. 13

(a) Barnes Ltd purchased equipment in 2021 for $80,000 and estimated an $8,000 salvage value at the end of the equipment's 10-year useful life. At 31 December, 2009, there was $50,400 in the Accumulated Depreciation account for this equipment using the straight-line method of depreciation. On March 31, 2030, the equipment was sold for $21,000.

Prepare the appropriate journal entries to remove the equipment from the books of Barnes Ltd on March 31, 2030.

(b) Lanne Manufacturing sold a delivery truck for $11,000. The delivery truck originally cost $25,000 in 2022 and $6,000 was spent on a major overhaul in 2027 (charged to Delivery Truck account). Accumulated Depreciation on the delivery truck to the date of disposal was $20,000.

Prepare the appropriate journal entry to record the disposition of the delivery truck.

(c) Crown Travel Ltd sold office equipment that had a book value of $4,500 for $6,000. The office equipment originally cost $15,000 and it is estimated that it would cost $19,000 to replace the office equipment.

Prepare the appropriate journal entry to record the disposition of the office equipment.

Solution 13 *(15 min.)*

(a)	Depreciation Expense	1,800	
	Accumulated Depreciation—Equipment		1,800
	(To record depreciation expense for the first 3 months of 2004. $7,200 x 1/4 = $1,800)		
	Cash	21,000	
	Loss on Disposal	6,800	
	Accumulated Depreciation—Equipment ($50,400 + $1,800)	52,200	
	Equipment		80,000
	(To record sale of equipment at a loss)		
(b)	Cash	11,000	
	Accumulated Depreciation—Delivery Truck	20,000	
	Delivery Truck		31,000
	(To record disposition on delivery truck at book value)		
(c)	Cash	6,000	
	Accumulated Depreciation—Office Equipment	10,500	
	Office Equipment		15,000
	Gain on Disposal		1,500
	(To record disposal of office equipment at a gain)		

Ex. 14

Prepare the journal entries to record the following transactions for the Nobles Company, which has a calendar year end and uses the straight-line method of depreciation.

(a) On 30 September, 2010, the company sold old delivery equipment for $9,000. The delivery equipment was purchased on 1 January, 2008, for $21,000 and was estimated to have a $3,000 salvage value at the end of its 5-year life. Depreciation on the delivery equipment has been recorded through 31 December, 2009.

(b) On 30 June, 2010, the company sold old office equipment for $18,000. The office equipment originally cost $24,000 and had accumulated depreciation to the date of disposal of $10,000.

Solution 14 *(15 min.)*

(a) 30 September, 2019		
Depreciation Expense	2,700	
Accumulated Depreciation — Delivery Equipment...........		2,700
(To record depreciation expense for the first 9 months of 2002. $18,000 / 5 years = $3,600 x 9/12 = $2,700)		
Cash	9,000	
Accumulated Depreciation — Delivery Equipment ($7,200 + $2,700)	9,900	
Loss on Disposal ($11,100 - $9,000)	2,100	
Delivery Equipment		21,000
(To record sale of delivery equipment at a loss)		
(b) 30 June, 2010		
Cash	18,000	
Accumulated Depreciation — Office Equipment	10,000	
Office Equipment		24,000
Gain on Disposal ($18,000 - $14,000)		4,000
(To record sale of office equipment at a gain)		

Ex. 15

a. A machine that cost $18,000 and on which $13,000 of depreciation had been recorded was disposed of for $5,200. Indicate whether a gain or loss should be recorded, and for what amount.
b. Assume that the machine in Part A above was instead discarded. Indicate whether a gain or loss should be recorded, and for what amount.
c. Assume that the machine in Part A above was instead sold for $4,800. Indicate whether a gain or loss should be recorded, and for what amount.

Solution 15 (12 min.)

a. $200 gain (18000 - 13,000 = 5000 book value; 5,200 - 5000 = 200 gain)
b. $5,000 loss (18000 - 13,000 = 5000 book value, all loss)
c. $200 loss (18000 - 13,000 = 5000 book value; 4,800 - 5000 = 200 loss)

Ex. 16

A patent that was acquired for $800,000 at the beginning of the current year expires in 20 years and is expected to have value for 4 years. Present the adjusting entry to amortise the patent for the current year.

Solution 16 (5 min.)

Amortisation expense — Patents……………………………	200,000	
Patents……………………………………………..		200,000

Ex. 19

(a) A company purchased a patent on 1 January, 2010, for $2,500,000. The patent's legal life is 20 years but the company estimates that the patent's useful life will only be 5 years from the date of acquisition. On 30 June, 2010, the company paid legal costs of $162,000 in successfully defending the patent in an infringement suit. Prepare the journal entry to amortise the patent at year end on 31 December, 2010.

(b) The Walker Company purchased a franchise from the Tasty Food Company for $400,000 on 1 January, 2010. The franchise is for an indefinite time period and gives the Walker Company the exclusive rights to sell Tasty Wings in a particular territory. Prepare the journal entry to record the acquisition of the franchise and any necessary adjusting entry at year end on 31 December, 2010. The franchise is expected to have value for a period of 20 years.

(c) Dawson Company incurred research and development costs of $500,000 in 2010 in developing a new product.

Instructions

Prepare the necessary journal entries during 2010 to record these events and any adjustments at year end on 31 December, 2010.

Solution 19 *(15 min.)*

(a) 31 December, 2010

Patent Expense		518,000	
Patent			518,000
(To record patent amortisation)			
$2,500,000 / 5 years	$500,000		
$162,000 / 54 months = $3,000 x 6	18,000		
	$518,000		

(b) 1 January, 2010

Franchise	400,000	
Cash		400,000
(To record acquisition of Tasty Food franchise)		

31 December, 2010

Franchise Expense ($400,000 / 20)	20,000	
Franchise		20,000
(To record amortisation of franchise for the current year)		

(c) 2010

Research and Development Expense	500,000	
Cash		500,000
(To record research and development expense for the current year)		